Certificate in Healthcare Infection Prevention & Control (CHIP)

Summer Internship

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Supervisor: Dr. Lona Mody
Location: Michigan Medicine - Center for Research and Innovations In Special Populations (CRIISP)
Internship Duration: May 6, 2024 - August 23, 2024

Internship Responsibilities

1. Assist in laboratory with microbiology testing and cultures as shown below;
2. Data entry and analysis of MDRO colonization, clinical data, antibiotic usage, and/or other Infection Control topics
3. Analyze MDRO colonization and contamination rates, assessing for trends and improvements, and providing monthly feedback reports to facilities
4. Assist in visits to post-acute care facilities with the field staff, including patient recruitment, specimen collection, medical record review and educational activities
5. Independent project development, including literature reviews, generating and testing hypotheses, creating tables and figures, and abstract and paper writing.

Individual Project: Persistence of MDROs in Long-term Care Facility (LTCF) setting

Background

- The persistence of multidrug-resistant organisms (MDROs) in healthcare settings is a critical global issue, with many aspects still unclear.
- Nursing homes (NHs) are increasingly responsible for post-acute care, making them key environments for managing MDROs. Veterans Affairs (VA) nursing homes (NHs), often located within hospitals, face unique challenges due to their proximity to acute care services.
- This study examines MDRO persistence, new acquisitions, and associated risk factors, focusing on multidrug-resistant gram-negative bacteria (RGNB), methicillin-resistant *Staphylococcus aureus* (MRSA), and vancomycin-resistant *enterococci* (VRE).
- We will also assess MDRO clearance and the time to clearance in patients with persistent colonization.

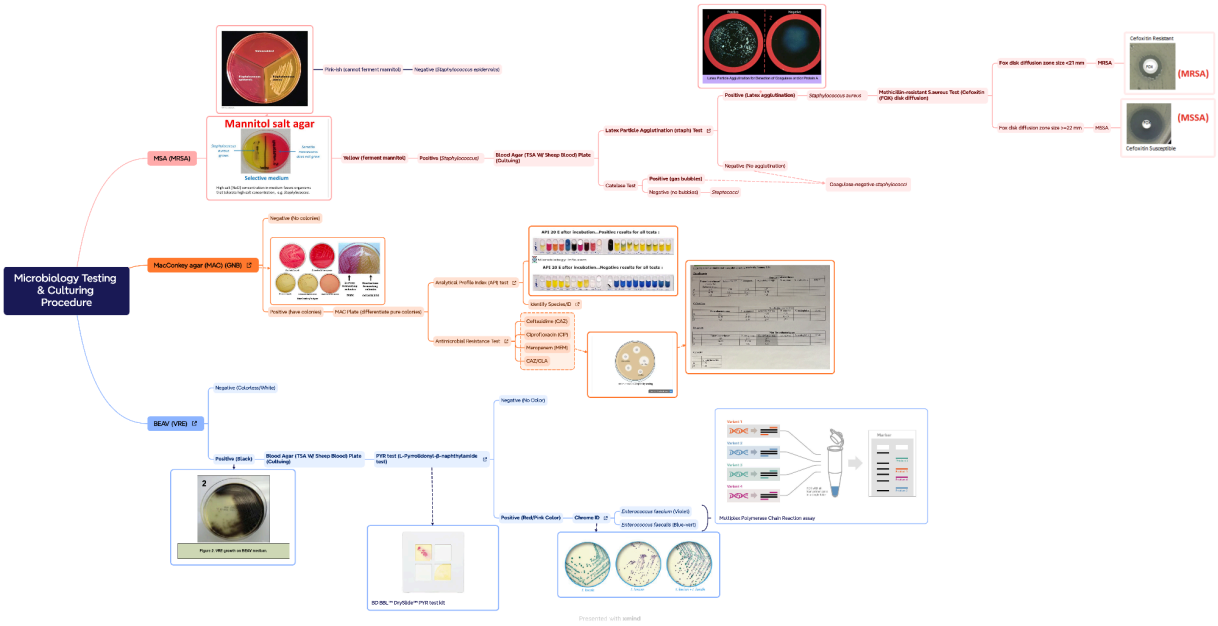
Methods

- Newly admitted Veterans at 3 VA NHs (April ‘21- Sept ‘23)
- Participants visited on day 0, 7, 14, 21, 30, 60, and 90.
- Samples collected from nares, groin, hands, and room surfaces, plus surfaces and hands at interactive visits (e.g., therapy, dialysis)

Microbiologic Methods

Samples were collected using sterile swabs in transport media (BactiSwab, Remel, Lenexa, Kansas) and cultured on mannitol salt agar, MacConkey agar, and bile-esculin plates with 6 µg/mL vancomycin. Hand swabs were enriched overnight in brain heart infusion broth at 36°C before culturing. Staphylococcus aureus, vancomycin-resistant *enterococci* (VRE), and gram-negative bacilli (GNB) were identified using

CDC standard microbiological testing[1]. Antimicrobial susceptibility testing for *S. aureus* and GNB isolates was performed by disk diffusion per Clinical and Laboratory Standards Institute guidelines, including screening for extended-spectrum β-lactamases (ESBLs)[2]. For *S. aureus*, methicillin resistance was determined by disk diffusion method using 30-µg cefoxitin disks. For GNB, the following antibiotics were tested: ceftazidime (30 µg), ceftazidime/clavulanic acid (30/10 µg), ciprofloxacin (5 µg), and meropenem (10 µg); resistance to 1 or more these was considered resistant GNB (R-GNB). We identified the species of VRE (ie, *Enterococcus faecalis*, *Enterococcus faecium*, other *enterococci*) using ChromID VRE medium (bioMérieux, Marcy l'Etoile, France) and the multiplex polymerase chain reaction assay[3]. Results of microbiological testing were not disclosed to NF staff or administrators during the study period.



(Clear version shows in **Microbiology Testing & Culturing Procedure.pdf**)

Definitions of Study Outcomes

1. Persistence and clearance of pre-existing colonization of MRSA, VRE, RGNB

- For residents with baseline positive culture, **intermittent** carriage is defined as two or more negative cultures after a single positive culture for any of the organisms. If there was no culture taken after a single positive culture, it was conservatively classified as an intermittent colonization. (i.e. no persistence)
- For residents with baseline positive culture, **persistence** is defined two or more positive cultures separated by fewer than two negative cultures
- For residents with MDRO persistence, **clearance** is indicated by two consecutive negative cultures
- For residents with MDRO persistence and clearance, **duration of persistence** is days from baseline visit to last positive culture immediately prior to clearance.
- For residents with MDRO persistence and NO clearance, **duration of persistence** is days from baseline visit to last positive culture.
- For any survival analysis, these residents should be treated as censored.

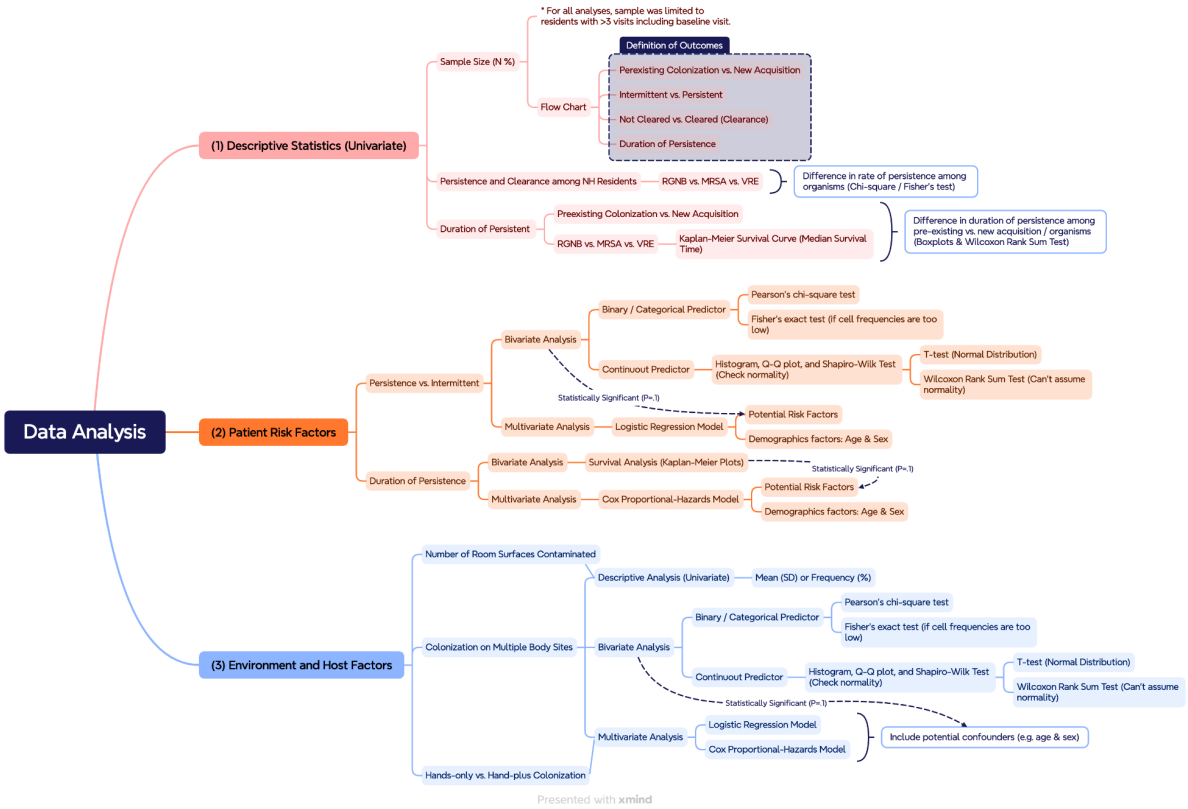
2. Persistence and clearance of new acquisition of MRSA, VRE, RGNB

- Definitions identical to outcome 1, with new acquisition culture used in place of the baseline positive culture.

Statistical Analysis

To evaluate factors potentially associated with persistent MRSA, VRE, RGNB carriage, for univariate analysis, continuous variables were compared using the nonparametric Wilcoxon rank-sum test and t-test, and categorical variables were compared using Pearson's chi-square test or Fisher's exact test, as appropriate. ORs and 95% CIs were calculated. Next, Multivariate analysis using conditional logistics regression was performed, in which variables were statistically significant (P=.1) in univariate models were entered into the multivariable model. All tests were 2-sided, and P values <.05 were considered significant. Statistical analysis was performed with use of R version 4.4.0 (RStudio, PBC, Boston, MA, USA)[4].

Duration of persistence was analyzed using similar methods, with tests selected based on the assumptions of normality and equal variances, including Welch's t-test, Mann-Whitney U test, One-Way ANOVA, Welch's ANOVA, or Kruskal-Wallis test for multiple-level categorical predictors.The duration of clearance of MRSA, VRE, RGNB, and any MDRO among NH residents were described using Kaplan-Meier estimates and actuarial life-table methods[5].



(Clear version shows in **MDRO Data Analysis.pdf)**

Reference

1. Centers for Disease Control and Prevention. Vital signs: stop spread of antibiotic resistance. https://www.cdc.gov/vitalsigns/pdf/2015-08-vitalsigns.pdf. Accessed 14 November 2017. [Ref list]
2. Ii L, S J, Institute C and LS. Performance standards for antimicrobial susceptibility testing. *(No Title)*. Accessed August 16, 2024. https://cir.nii.ac.jp/crid/1130578271562670753
3. Dutka-Malen S, Evers S, Courvalin P. Detection of glycopeptide resistance genotypes and identification to the species level of clinically relevant enterococci by PCR. *J Clin Microbiol*. 1995;33(1):24-27. doi:10.1128/jcm.33.1.24-27.1995
4. RStudio Team. RStudio: Integrated Development for R. RStudio, PBC, Boston, MA. http://www.rstudio.com/
5. Kaplan EL, Meier P. Nonparametric Estimation from Incomplete Observations. *Journal of the American Statistical Association*. 1958;53(282):457-481. doi:10.1080/01621459.1958.10501452